MASTER OF SCIENCE IN DATA ANALYTICS

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION SYSTEMS

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Enhancing Customer Data Quality and Integration in Financial Institutions: A Data Analytics Approach

The main focus of this study is to provide the practical solutions to real world issues through join two methods namely qualitative and quantitative. To achieve this approach, the study will adopt a pragmatic epistemology. This approach to knowledge is suitable for this study because data quality and integration problems in the financial institutions depends on measurable data driven views and an understand of organization practices controlling these issues.

For this study the financial institution we are looking into is old mutual Malawi. For this company data is generated across several systems to produce reports that have to be developed into one. Because these systems often collect similar information and work in silos these develops duplication and data inconsistence which strongly affect the accuracy and decision making of the reporting. Using the pragmatic approach will allow flexibility to explore this complex issue by joining qualitative approach such as data profiling and accurate measurement with insights from ict managers and staff.

Pragmatism values both positivist stance as well as interpretivist methods. Positivist stance solely depends on numerical data while an interpretivist approach solely depends only on meaning and experience. Using the pragmatism for this study with allow us to use both, it will also allow the researcher to develop technical data analytics with generic understanding.

2. Conceptualisation and Operationalisation

Key Concepts and Definitions

1. **Customer Data Quality**
   This can be explained as how customer data is direct; timely, complete, consistent and legitimate. Analytics, customer satisfaction, and regulatory compliance are all contained by poor data quality.
   - *Operationalisation:* Measured through quantitative pointers such as completeness percentage, duplication rate, accurate ratio, and timeliness index.
2. **Data Integration**
   This is the process of combining information from different systems into a single. Integration grantees data consistency across platforms and departments.
   - *Operationalisation:* Measured by integration latency , consistency between systems, and the number of duplicate records identified.
3. **Data Governance**
   A structured framework defining roles, policies, and standards for managing data across its lifecycle (Otto, 2021).
   - *Operationalisation:* Evaluated through policy compliance levels, employee awareness of data standards, and frequency of governance audits.
4. **Data Analytics**
   The use of computational and statistical techniques to extract useful insights from data (Madnick et al., 2019).
   - *Operationalisation:* Measured by the number of data anomalies detected, profiling accuracy, and performance of cleansing algorithms

Both quantitative measurements and qualitative assessments will be utilized to create these ideas. Interviews will provide views into governance and system development procedures, while data profiling and visualization will quantify data issues.

3. Problem Statement and Research Objectives

Problem Statement

Data-driven decision-making is important for financial institutions, but its importance is usually not valued by system fragmentation and poor data. Old Mutual Malawi manages customer data in a number of databases for claims, policies, and customer service. This leads to incomplete records, missing information, and a lag in system synchronization. These issues minimize the accuracy of analytics results, affect customer profiling, and complicate regulatory compliance.

The upcoming Malawi Data Protection Bill raises the need for accurate, consistent, and well-managed data. Few Malawian financial institutions, nevertheless, have implemented analytics-based systems for continuous data quality tracking. The development of a structured, data-driven framework that improves data quality and integration is necessary to support compliance, efficiency in operation, and informed choices.

Background and Literature Review

A key element of analytics levels and digital transformation is high-quality data. Redman (2018) addresses that customer satisfaction, risk management, and business intelligence are all directly impacted by data quality. Batini et al. (2009) explained that by providing trustworthy datasets for analysis, systematic data quality assessment raises the accurate of decision-making. While Cichy and Rass (2019) in directed that data governance frameworks that are well-structured guarantee data trustworthiness, accountability, and consistency across systems.

According to Madnick et al. (2019), financial services organizations' scale to develop a complex picture of their customers is affected by systems that are not connected and inadequate data development. According to Otto (2021), organizations find it hard to align data sources in the absence of effective governance, which results in missing or duplicate information. Ngulube (2020) points that these problems are made worse in developing nations by a lack of technical education, a poor data management culture, and low ICT infrastructure.

Data analytics approaches like data profiling, cleansing algorithms, and integration pipelines can be used to evaluate and improve the quality of customer data in order to close these gaps. There is not much empirical research, though, that is important to the financial environment in Malawi.

Research Objectives

1. To assess the current state of customer data quality and integration at Old Mutual Malawi.
2. To identify key factors affecting data accuracy, completeness, and consistency.
3. To develop a data analytics framework for enhancing customer data quality and integration.
4. To recommend ICT-based strategies for sustainable data governance and compliance.

4. Research Methods

Research Design

The study will employ a **mixed-methods design**, joining both quantitative and qualitative methods. Quantitative analysis will identify measurable data quality problems, while qualitative interviews will access organizational and governance elements contributing to those problems. This combination aligns with the pragmatic epistemology, which values both measurable evidence and contextual understanding.

Sampling Strategy

- **Quantitative Sample:** A simulated dataset of 5,000 anonymized customer records drawn from multiple systems (CRM, policy, and claims).
- **Qualitative Sample:** A purposive sample of 10–12 ICT and data management staff from Old Mutual Malawi, including system administrators, compliance officers, and data analysts.

Data Collection Methods

- **Quantitative:**
  - Secondary or simulated customer data will be used.
  - Data profiling tools such as **Python (pandas profiling)** and **SQL queries** will compute completeness, duplication, and accuracy metrics.
  - Visualization platforms like **Power BI** will present trends and integration performance.
- **Qualitative:**
  - Semi-structured interviews will be conducted to understand system-level and policy-related challenges.
  - Company policy documents and data governance reports will also be reviewed.

Data Analysis Techniques

- **Quantitative Analysis:**
  Descriptive statistics and correlation analysis will identify data quality patterns and relationships between data attributes.
- **Qualitative Analysis:**
  Thematic analysis will be used to code and interpret interview transcripts, focusing on governance practices, human errors, and integration gaps.

By combining these methods, the study will provide both empirical and interpretive insights into how data analytics can improve information quality and system integration.

5. Expected Results

It is expected that the study will undercover widespread data quality problems such as incomplete fields, duplication of data, inconsistent updates across systems, and low synchronization between databases. Organizational elements such as poor governance policies, low training, and lack of centralized oversight are expected to be identified as key contributors.

The proposed **data analytics framework** will likely demonstrate how continuous data profiling, automated cleansing, and dashboard-based monitoring can improve the completeness and reliability of customer data.

Expected Contributions

- **Theoretical Contribution:** Extends the literature on data quality and governance by developing an analytics-based framework suited for financial institutions in developing economies.
- **Practical Contribution:** Provides Old Mutual Malawi with actionable recommendations for improving data management practices and achieving compliance with regulatory standards.
- **Policy Contribution:** Supports policymakers in strengthening Malawi's Data Protection Bill implementation through evidence-based data governance strategies.

Overall, the study aims to bridge the gap between academic research and ICT practice by showing how data analytics can directly improve institutional data governance and customer service delivery.

6. Ethics

Ethical conduct is important in research focusing financial data. The study will include both **university research ethics protocols** and **Old Mutual's internal data protection policies**.

- **Data Privacy:** Only simulated or anonymized data will be used to ensure no customer data is exposed.
- **Informed Consent:** Written consent will be obtained from all participants and from Old Mutual management before conducting interviews or accessing internal materials.
- **Confidentiality:** Responses and internal documents will be securely stored and not shared beyond the research scope.
- **Data Security:** All electronic data will be stored on password-protected and encrypted systems, with backups in secure cloud storage.
- **Compliance:** The study will follow Malawi's forthcoming **Data Protection Bill** and align with international standards such as the **General Data Protection Regulation (GDPR)**.

7. Conclusion

This study proposal has explained a complete plan to examine how data analytics can improve customer data quality and development within financial institutions, using **Old Mutual Malawi** as a case study. Unitizing a **pragmatic epistemology**, the study develops both quantitative and qualitative methods to provide a balanced understanding of the technical and organizational factors that affect data management.

By conceptualising and operationalising key constructs such as **data quality**, **integration**, **governance**, and **analytics**, the proposal establishes a strong theoretical foundation for empirical investigation. The proposed mixed-methods design promotes that measurable views from data profiling are complemented by contextual understanding from staff interviews.

The expected advantages will advance data governance theory and policy in addition to offering a workable framework for ongoing data quality improvement. The ultimate goals of this study are to improve data-driven decision-making, support adherence to new data protection regulations, and fortify ICT practices in Malawi's financial sector.

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